SCHOOL OF ELECTRONICS ENGINEERING (SENSE)

About School

The Department of Electronics and Communication Engineering was established in 2017 and became the School of Electronics Engineering (SENSE) in January 2019. At present, the School is headed by Dr. Umakanta Nanda. The School has qualified and committed faculty members who offer impressive instruction in a variety of modes, exploring experiential learning approaches and new pedagogical methods in order to provide proper learning to students. The School provides the necessary infrastructure, hardware, and software to support the faculty for their continuous learning and research. The programmes offered under SENSE are meticulously designed for a strong real-world application and industrial relevance. They range from the application of science in transforming semi-conductor materials as usable devices, which may be used to create "smart" innovative products, to enabling students design efficient, affordable, means and mechanisms for communicating information. At present, the following programmes are offered at the undergraduate and post-graduate level.

Eligibility

Nationality

- The applicant should be a Resident / Non-Resident Indian National / PIO / OCI.
- NRI applicants can directly apply under 'NRI Category' through the NRI application form.
- Foreign applicants who have studied/are studying abroad can apply directly through the International application form.
- Visit https://vit.ac.in/admissions/international/overview to know more.

Age Limit

- Applicants whose date of birth falls on or after 1st July 2002 are eligible to apply for admission into Engineering in the academic year 2024.
- The date of birth as recorded in the High School / SSC / X Certificate will be considered authentic.
- Applicants should produce this certificate in original as a proof of their age at the time of counselling/admission, failing which their candidacy for admission will be *cancelled*.

UNDERGRADUATE

ECE

About the Programme

The objective of this programme is to learn the concepts of electronics and communication disciplines and apply them to solve real world problems in the industry. It emphasizes the design of

hardware systems as well as communication systems. The students start the programme learning the fundamental concepts of Electrical Engineering, while also studying foundation courses in Mathematics and Physics in tandem. A student, then, progresses to study core subjects of Electronics, such as Analog and Digital Circuit Design, Control Engineering, and communications courses, such as Analog and Digital Communications, Signals and Systems, etc. The foundations laid by these basic courses enable the students to select various electives that provide them with specialized skill sets.

Career Prospects

A student successfully completing this programme would be well equipped with necessary skills to either pursue higher studies or work in industry as IC design engineer, Systems engineer, Embedded Systems Design engineer, RF engineer, Wireless/Mobile network engineer etc.

Design Your Own Degree (DYOD)

- Students are given options to select either a Specialization or a Minor along with their choice of B.Tech. Major during counselling.
- A student is allowed to change his/her choice of Specialization or Minor after the completion of their first year in the University as per their evolving interests.
- A student can also choose to do a Double Major instead of a Specialization or a Minor when they start their second year.

Specialization

- VLSI Design and Embedded systems.

Minors

- Artificial Intelligence and Machine Learning
- Data Analytics
- Cyber Security
- Edge Computing
- Geo Informatics
- Software Engineering
- Blockchain
- Computer Science & Engg.
- Digital Manufacturing
- Automotive Design
- Robotics
- Mechanical Engineering
- FinTech
- Digital Marketing
- Econometrics
- Performing Arts
- Computational Mathematics
- Corporate Law

Double Major

- Computer Science and Engineering
- Mechanical Engineering
- Applied Science
- Business Management
- Economics

Program Outcomes

After successful completion of the program a student is expected to have abilities to

PO1. Engineering knowledge:

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis:

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions:

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems:

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage:

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society:

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability:

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics:

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work:

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication:

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance:

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning:

Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Educational Objectives

- *Core competency :* To apply fundamentals of Electronics and Communication Engineering in their professional practice and become successful engineers, innovators, consultants, managers, researchers, or entrepreneurs.
- *Professional and responsible competency :* To use methodologies of electronic system design, analysis, optimization, and development considering social and industrial needs.
- *Administrative competency :* To update and apply the knowledge in administrative skills through higher education, professional training, and courses ensuring a better future.

Programme Specific Outcome

- To utilize the basics and fundamentals of Electronics and Communication Engineering for theoretical developments of complex problems.
- To perform signal processing methodologies to develop communication systems and system components.
- To learn adequate tool and techniques for improving the throughput, security and efficiency of the communication systems by updating the knowledge base through various mediums.

ECE-EMBEDDED-SYSTEMS

About the Programme

The objective of this specialization or minor is to provide the key aspects of both hardware and software design and development for real time applications. The prime focus is laid on niche areas such as, Embedded Controllers, Real Time Operating Systems, Automotive Electronics and In-Vehicle Networking, Agriculture, and Health Care. The embedded discipline is an intersection of Computer Science and Electrical Engineering and its applications are found in sectors such as Automotive Industry, Avionics, Consumer Electronics, Medical Devices and Defence.

Career Prospects

A career specialization in Embedded Systems, as part of a B.Tech programme, opens up various opportunities in the rapidly evolving field of technology. Embedded Systems are integral to a wide range of industries, and professionals with expertise in this area are in high demand. Most of the leading companies such as Texas Instruments, NXP, Wipro, GE, Motorola, Honeywell, VOLVO, PHILIPS, HCL Technologies, and TATA are recruiting hardware and software system designers to involve in their R&D.

Design Your Own Degree (DYOD)

Students are given options to select either a Specialization or a Minor along with their choice of B.Tech. Major during counselling.

A student is allowed to change his/her choice of Specialization or Minor after the completion of their first year in the University as per their evolving interests.

A student can also choose to do a Double Major instead of a Specialization or a Minor when they start their second year.

Specialization

- VLSI
- Image and Video Processing
- Embedded Systems
- Internet of Things (IoT)
- Electric Vehicle Technology

Minors

- Artificial Intelligence and Machine Learning
- Cyber Security
- Edge Computing
- Geo Informatics
- Software Engineering
- Blockchain
- Robotics
- Mechanical Engineering

- FinTech
- Digital Marketing
- Econometrics
- Computational Mathematics
- Corporate Law

Double Major

- Computer Science and Engineering
- Mechanical Engineering
- Business Management

Program Outcomes

After successful completion of the program a student is expected to have abilities to

PO1. Engineering knowledge:

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis:

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions:

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems:

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage:

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society:

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability:

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics:

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work:

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication:

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance:

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning:

Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Educational Objectives

- *Core competency* : To apply fundamentals of Electronics and Communication Engineering in their professional practice and become successful engineers, innovators, consultants, managers, researchers, or entrepreneurs.
- *Professional and responsible competency* : To use methodologies of electronic system design, analysis, optimization, and development considering social and industrial needs.
- *Administrative competency :* To update and apply the knowledge in administrative skills through higher education, professional training, and courses ensuring a better future.

Programme Specific Outcome

- To utilize the basics and fundamentals of Electronics and Communication Engineering for theoretical developments of complex problems.
- To perform signal processing methodologies to develop communication systems and system components.
- To learn adequate tool and techniques for improving the throughput, security and efficiency of the communication systems by updating the knowledge base through various mediums.

ECE-VLSI

About the Programme

The objective of this programme, honours as well as minor, is to provide a student with comprehensive knowledge of VLSI circuits and systems, which is essential for electronics chip manufacturing industry. The programme emphasizes the key aspects of hardware design and development for VLSI applications. The prime focus is laid on the areas such as VLSI system design, verification methodologies, ASIC design, FPGA based systems design, and SOC based design and verification.

Career Prospects

The VLSI field is a highly specialized vocation that opens doors to individuals with technical expertise in the areas of chip fabrication, processor design, and their verification and testing. As our day to day life devices are getting smarter, smaller, and still being user friendly and reliable, the VLSI Engineers have an ace over others. Industries such as Qualcomm, Samsung, Dell, Apex Semiconductors, Intel, PHILIPS, NXP, Texas Instruments, ARM and others are the major recruiters for the students with this either as a specialization or as a minor.

Design Your Own Degree (DYOD)

Students are given options to select either a Specialization or a Minor along with their choice of B.Tech. Major during counselling.

A student is allowed to change his/her choice of Specialization or Minor after the completion of their first year in the University as per their evolving interests.

A student can also choose to do a Double Major instead of a Specialization or a Minor when they start their second year.

Specialization

- VLSI
- Image and Video Processing
- Embedded Systems
- Internet of Things (IoT)
- Electric Vehicle Technology

Minors

- Artificial Intelligence and Machine Learning
- Data Analytics
- Cyber Security

- Edge Computing
- Geo Informatics
- Software Engineering
- Blockchain
- Computer Science & Engg.
- Digital Manufacturing
- Automotive Design
- Robotics
- Mechanical Engineering
- FinTech
- Digital Marketing
- Econometrics
- Performing Arts
- Computational Mathematics
- Corporate Law

Double Major

- Computer Science and Engineering
- Mechanical Engineering
- Applied Science
- Business Management
- Economics

Program Outcomes

After successful completion of the program a student is expected to have abilities to

PO1. Engineering knowledge:

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis:

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions:

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4.Conduct investigations of complex problems:

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5.Modern tool usage:

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6.The engineer and society:

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7.Environment and sustainability:

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development

PO8. Ethics:

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and teamwork:

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication:

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance:

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning:

Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Educational Objectives

- *Core competency:* To apply fundamentals of Electronics and Communication Engineering in their professional practice and become successful engineers, innovators, consultants, managers, researchers, or entrepreneurs.
- *Professional and responsible competency:* To use methodologies of VLSI system design, analysis, optimization and development considering social and industrial needs.

- *Administrative Competency:* To update and apply the knowledge in administrative skills through higher education, professional training and courses ensuring a better future.

Programme Specific Outcome

- To use programming languages and software for projects related to VLSI Physical Design, and ASIC Design or verification and become industry ready.
- Based on the research experience gained from semiconductor technologies, they will carry out VLSI Design verification and testing, SOC design and FPGA based systems.

POSTGRADUATE

VLSI

About the Programme

The post-graduate programme in VLSI Design at VIT-AP University aims to provide fundamental concepts related to the design of VLSI circuits and systems. Students in the programme are trained to understand the complete VLSI design flow using various industry-standard Electronic Design Automation (EDA) tools like Synopsys and Cadence. The curriculum of VLSI Design has been designed to address the dynamic demands of the VLSI industry and the evolving needs of society through cutting-edge research. The focus areas of the programme include digital system and circuit design, analog design, physical design, verification methodologies, scripting languages, and semiconductor device modeling.

Highlights

- **Core Courses:**

  Digital System Design for ASICs and FPGA, Analog System Design, CMOS VLSI Design, HDL Verification Methodologies, SoC Design, Physical Design.

- **Electives:**

  Semiconductor Device Modeling, Low Power IC Design, VLSI Testing and Testability, Scripting Languages and Verification, VLSI Digital Signal Processing, Computer-Aided Design for VLSI, Memory Design and Testing, Mixed Signal IC Design, RFIC Design.

- **IC Design Lab and ASIC Design Lab:**

  Fully equipped with industry-standard EDA Tools with 40 user licenses and EDA Tools from Cadence. The lab also facilitates 5 user licenses for Synopsys TCAD, allowing students and researchers to gain hands-on experience in both analog and digital IC design, from front end to GDSII.

- **FPGA/SOPC Lab:**

Equipped with the ALTERA Design Suite and Xilinx Programmable Logic Design Tools Vivado software with 60 user licenses. The lab also features hardware boards such as Cyclone DE2 Boards from Altera and Xilinx ZYNQ Boards.

Career Prospects

- Top Recruiting Companies: Graduates are recruited by leading companies such as Intel, AMD, Qualcomm, Synopsys, Texas Instruments, NXP Semiconductors, Broadcom, Cadence Design Systems, MediaTek, and Infineon Technologies, with attractive starting salaries and significant growth potential.
- Highly Placement Rate and Competitive Packages: Graduates of previous batches are placed with registered student's placement rate of 90% and with Competitive Packages reaching up to 34 LPA.
- Diverse Job Profiles: Roles include VLSI Design Engineer, ASIC Design Engineer, FPGA Design Engineer, Verification Engineer, and more.
- Core Responsibilities: Involves designing and implementing integrated circuits, conducting research and development for semiconductor technologies, and collaborating with cross-functional teams.
- Wide Career Opportunities: Available in semiconductor, consumer electronics, telecommunications, automotive electronics, and medical devices industries. Most of the previous batch students are placed in renowned VLSI companies, showcasing the strong career support provided by the programme.

Program Outcomes

After successful completion of the program a student is expected to have abilities to

PO1. Modern tool usage:

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO2. The engineer and society:

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO3. Environment and sustainability:

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO4. Ethics:

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO5. Individual and team work:

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO6. Communication:

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO7. Project management and finance:

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO8. Life-long learning:

Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PO9. Conduct investigations of complex problems:

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO10. Design/development of solutions:

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO11. Problem analysis:

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO12. Engineering knowledge:

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

Programme Educational Objectives

- Core Competency: To impart engineering skills in professional practice and become successful engineers, innovators, consultants, managers, researchers or entrepreneurs.
- Professional and responsible competency: To work in the VLSI profession with social awareness and responsibility by interacting with their peers in other disciplines in industry contributing to the economic growth of the country.
- Administrative competency: To pursue higher studies in engineering, management for career paths in industry or teaching and research.

Programme Specific Outcome

- To design ASIC and FPGA based circuits and systems using industry standard programming languages and tools.
- To design and apply various concepts of system on chip, verification methodologies, design for testability and able to do mini projects.
- To identify research gaps and provide solutions to socio-economic, and environmental problems

Ph.D. Program

Overview

The Department of Electronics and Communication Engineering was established in 2017. It has become the School of Electronics Engineering (SENSE) in January 2019. At present, the School is headed by Dr. Umakanta Nanda with 859 students, 63 faculty and 144 research scholars. The School has qualified and committed faculty members who offer impressive instruction in a variety of modes, exploring experiential learning approaches and new pedagogical methods in order to provide proper learning to students. The school provides the necessary infrastructure, hardware, and software to support the faculty for their continuous learning and research. The programmes offered under SENSE are meticulously designed with a strong real-world application and industrial relevance. They range from the application of science in transforming semi-conductor materials as usable devices which may be used to create “smart” innovative products, to enabling students design efficient, affordable, means and mechanisms of communicating information. At present, the following programmes are offered at the undergraduate level and post-graduation level.

Selection Procedure for International/NRI candidates

Foreign nationals / NRI candidates can only register as full-time scholars through separate link:

https://admissions.vit.ac.in/irapplicationpg/

Major Areas of Research

- Antennas and Microwave Engineering:

  Our research in this domain delves into the design, analysis, and optimization of antennas and microwave systems. We explore advanced antenna technologies, microwave circuits, and propagation phenomena crucial for modern communication systems and radar applications.

- Communication Systems:

  Our research encompasses the design, analysis, and optimization of communication systems spanning various domains such as wireless, optical, and satellite communications. We focus on developing innovative solutions to enhance data transmission, networking protocols, and spectrum utilization efficiency.

- VLSI and Embedded System Design:

  In this field, our research focuses on the design, fabrication, and testing of integrated circuits and embedded systems. We explore techniques to improve system performance, energy efficiency, and reliability for applications ranging from consumer electronics to industrial automation.

- Signal Processing:

  Our research in signal processing involves the development of algorithms and methodologies for analyzing, manipulating, and interpreting signals. We address challenges in areas such as image processing, audio processing, and biomedical signal analysis to extract meaningful information and enhance system performance.

- Artificial Intelligence and Machine Learning:

  Our research in this area focuses on developing algorithms, models, and systems that can exhibit intelligent behavior, learn from data, and make decisions. This includes applications in areas such as natural language processing, computer vision, robotics, and autonomous systems.

- Control and Automation:

  Our research explores the theory and applications of control systems and automation techniques. We investigate methodologies for modeling, analyzing, and controlling dynamic systems, with applications in robotics, manufacturing, and process control.

- Power Electronics and Power Systems:

  Research in this domain focuses on the design, analysis, and optimization of power electronic converters and systems. We explore advanced power semiconductor devices, converter topologies, and control strategies for efficient energy conversion and management in various power system applications.

- Renewable Energy:

  Our research addresses challenges in renewable energy generation, integration, and management. We explore technologies such as solar photovoltaics, wind energy, and energy storage systems to enhance the sustainability and reliability of power supply.

- Electrical Vehicles:

  In this field, our research focuses on electric vehicle propulsion systems, charging infrastructure, and energy management strategies. We investigate technologies to improve the performance, efficiency, and environmental impact of electric vehicles for sustainable transportation.

- Battery Management Systems:

  Systems: Our research in battery management systems involves the development of algorithms and control strategies for monitoring, balancing, and optimizing the performance of battery packs. We address challenges in battery safety, longevity, and energy efficiency for various applications including electric vehicles and renewable energy storage.

- Image and Speech Processing:

Our research focuses on developing algorithms and systems for processing and analyzing images, videos, and speech signals. We explore techniques for image recognition, video understanding, and speech synthesis to enable applications in multimedia, healthcare, and human-computer interaction.